INSIGHTS DOCUMENT

Brain Tumor MRI Image Classification

Evidence-based findings on data quality, model behaviour and deployment readiness

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Domain	Medical imaging - deep learning image classification
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Live application	Streamlit app
Repository	github.com/SUMANSANGEET/Brain-Tumor-MRI-Classification

Research prototype - not a medical device. All results come from a single public dataset and must not be used for clinical decisions. Figures in this document are taken from the project notebook's executed outputs.

Executive snapshot

87.5%	0.864	0.976	65%
Test accuracy	Macro-F1	ROC-AUC (OvR)	Weakest-class recall (meningioma)

The project classifies brain MRI slices into glioma, meningioma, pituitary tumor and no tumor. The selected model, a fine-tuned ImageNet **ResNet50V2**, scores 87.5% accuracy on a 407-image test set that is free of source-scan leakage. The most valuable findings are not the headline score, however, but three things the analysis uncovered along the way: the vendor's train/test split leaked, class-level brightness differences offer a shortcut a network could exploit, and meningioma remains the class that needs the most attention before any real-world use. Each insight below follows an **Observation - Evidence - Implication** structure.

A. Data insights

D1 The dataset is only mildly imbalanced	
Observation	Glioma is the largest class and no-tumor the smallest.
Evidence	805 glioma (32.9%), 610 pituitary (25.0%), 545 meningioma (22.3%), 483 no tumor (19.8%); imbalance ratio 1.67:1. Vendor split proportions are statistically consistent (chi-square p = 0.47).
Implication	Inverse-frequency class weights are sufficient; resampling is unnecessary. Macro-averaged metrics should be used so the smallest class stays visible.

D2 The vendor's train/test split leaks	
Observation	Augmented copies of the same scan sit in different folders.
Evidence	352 source scans appear twice; 164 source IDs span more than one split; 67 of the 246 vendor test images (27%) have a sibling in train or valid. No byte-identical duplicates were found, so the problem is invisible to a naive MD5 check.
Implication	Scores on the vendor split would be inflated. A StratifiedGroupKFold re-split by source ID (1,629 / 407 / 407, zero overlap asserted in code) gives an honest estimate.

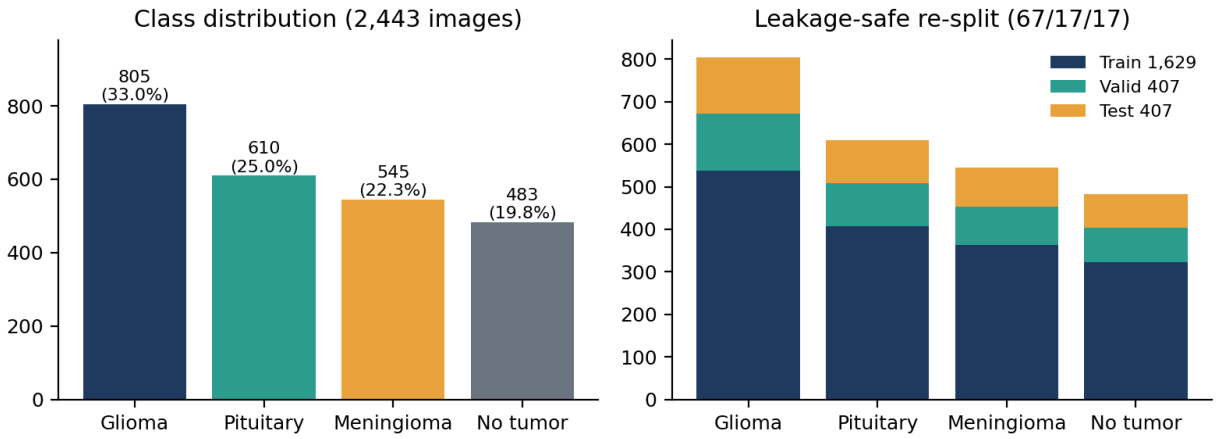


Figure 1 - Class balance (left) and the leakage-safe re-split (right).

D3 Classes carry different acquisition signatures	
Observation	Global image statistics separate the classes before any CNN is trained.
Evidence	Mean brightness: glioma ~28, meningioma ~46, pituitary ~50, no tumor ~60; foreground coverage 0.41 / 0.51 / 0.66 / 0.61. Kruskal-Wallis and ANOVA reject equality ($p < 1e-100$). A Random Forest on five simple descriptors (brightness, contrast, foreground ratio, sharpness, file size) reaches 80.7% accuracy and 0.800 macro-F1 under grouped cross-validation.
Implication	A network could score well by reading framing and intensity instead of pathology. The deep models beat this baseline by only 6 macro-F1 points (0.864 vs 0.800), so Grad-CAM review of attention is essential, not optional.

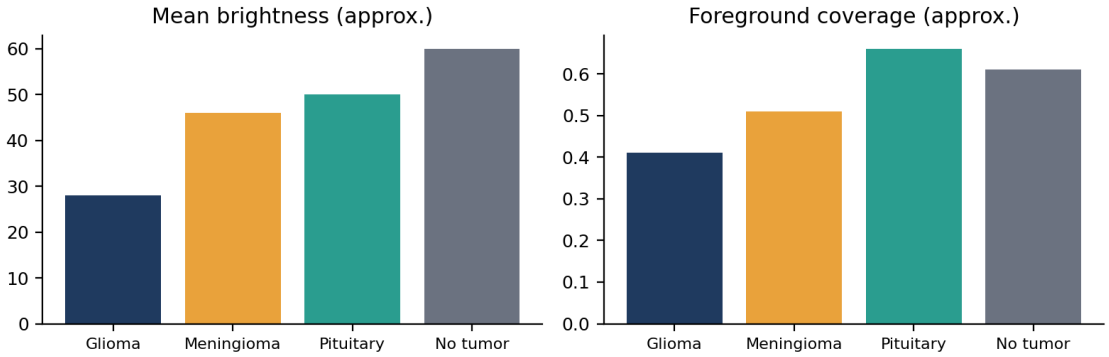


Figure 2 - Class means of two global descriptors (values as reported in the notebook's EDA, rounded).

D4 Images are RGB-encoded but effectively grayscale	
Observation	All 2,443 images are 640 x 640 JPEGs; the mean R-G difference is about zero.
Evidence	No missing metadata, no unreadable files, uniform size (median file 30 KB, 74.4 MB total).
Implication	ImageNet backbones can be used directly after resizing to 224 x 224; no cropping or padding logic is needed.

B. Model insights

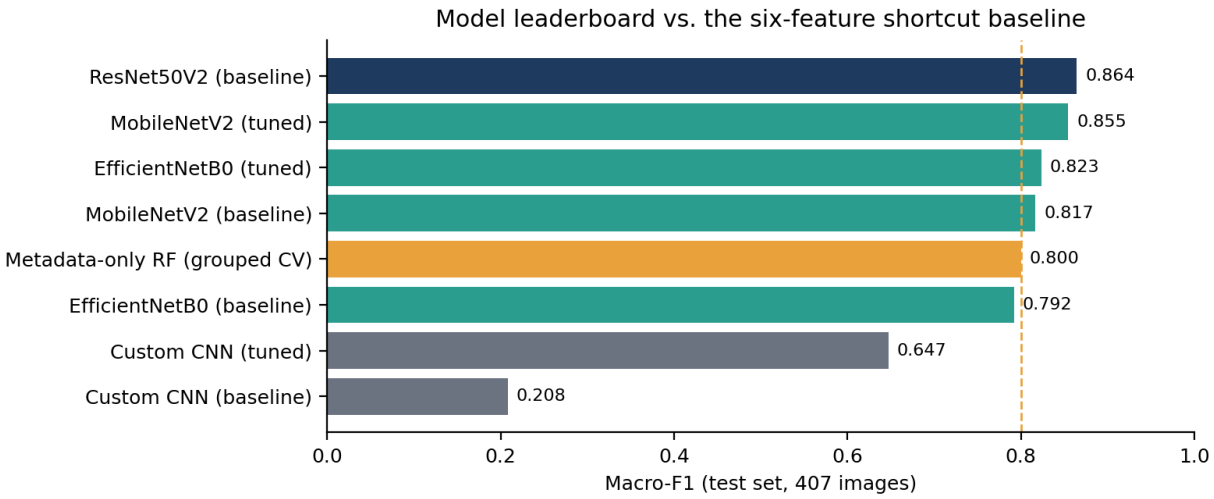


Figure 3 - Test macro-F1 for every model and stage, with the metadata-only baseline for context.

M1 Transfer learning beats training from scratch with ~1,600 images	
Observation	Every pretrained backbone outperforms the custom CNN by a wide margin.
Evidence	Best custom CNN: 0.647 macro-F1 (67.3% accuracy) after tuning, versus 0.864 for ResNet50V2. The untuned custom CNN reached only 0.208 macro-F1; its three-fold CV scores swung from 0.313 to 0.671.
Implication	Pretrained features are the right default for small medical datasets. The from-scratch network is an instructive baseline but not a deployment candidate.

M2 Tuning helps, but the gain depends on the starting point	
Observation	Cross-validated head search plus top-layer fine-tuning improved every model that was tuned.
Evidence	Macro-F1 baseline to tuned: custom CNN 0.208 to 0.647; MobileNetV2 0.817 to 0.855 (+3.8 pp); EfficientNetB0 0.792 to 0.823 (+3.2 pp). Cached-embedding search reached CV macro-F1 of 0.907 (MobileNetV2) and 0.910 (EfficientNetB0).
Implication	Tuning is worth the effort for lightweight backbones. Note that ResNet50V2 appears in the leaderboard at its baseline stage only, so it has not yet had the same tuning treatment.

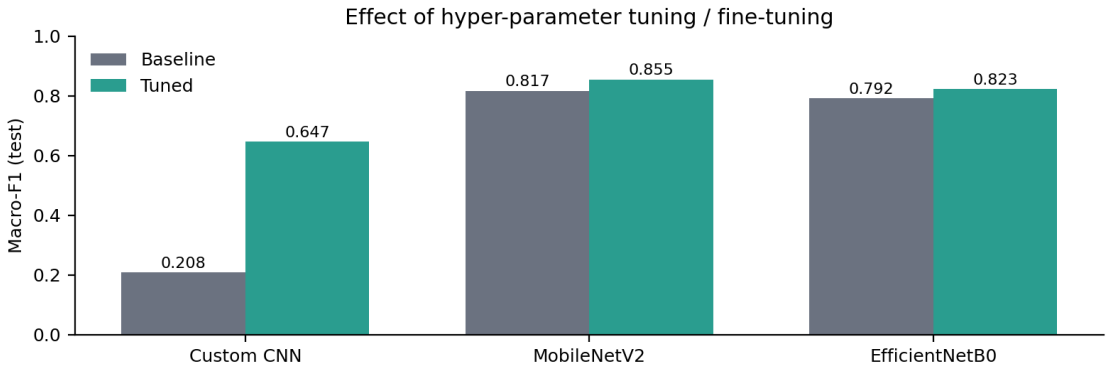


Figure 4 - Baseline versus tuned macro-F1.

M3 A lighter model is nearly as good	
Observation	MobileNetV2 (tuned) is within one macro-F1 point of the winner at a fraction of the cost.
Evidence	ResNet50V2: 0.864 macro-F1, 24.1 M parameters, 112 ms per image. MobileNetV2 tuned: 0.855 macro-F1, 2.9 M parameters, 50 ms. That is 8.3x fewer parameters and 2.2x faster for a 0.9 pp macro-F1 loss. ResNet50V2 was selected because its validation macro-F1 (0.888 vs 0.881) exceeded the 0.5 pp tie-break window.
Implication	For a CPU-hosted web app or low-resource clinic, MobileNetV2 is a credible alternative. The choice should be revisited once external validation shows which model generalises better.

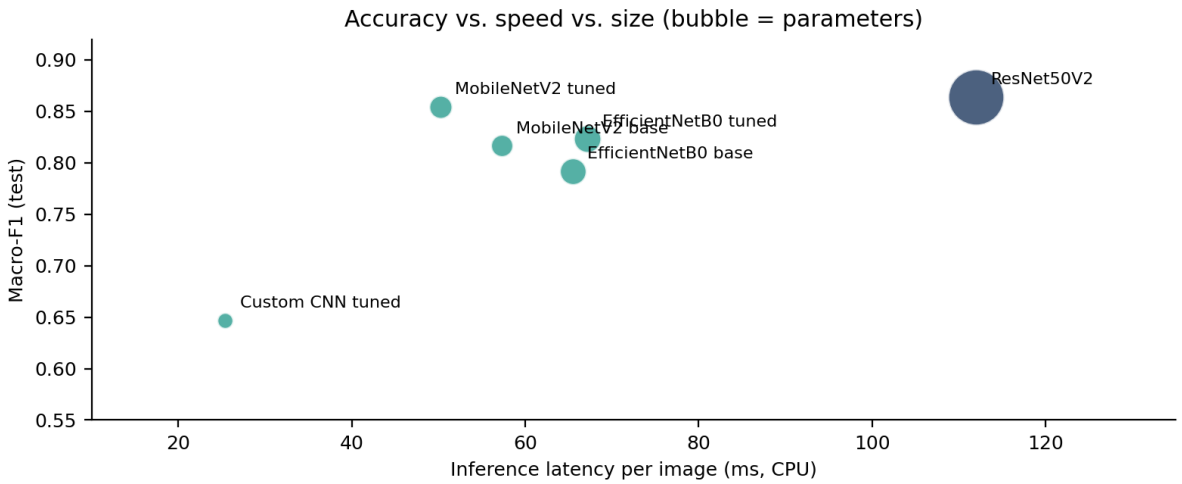


Figure 5 - Accuracy, latency and size trade-off (CPU inference, as measured in the notebook).

C. Class-level insights (final model)

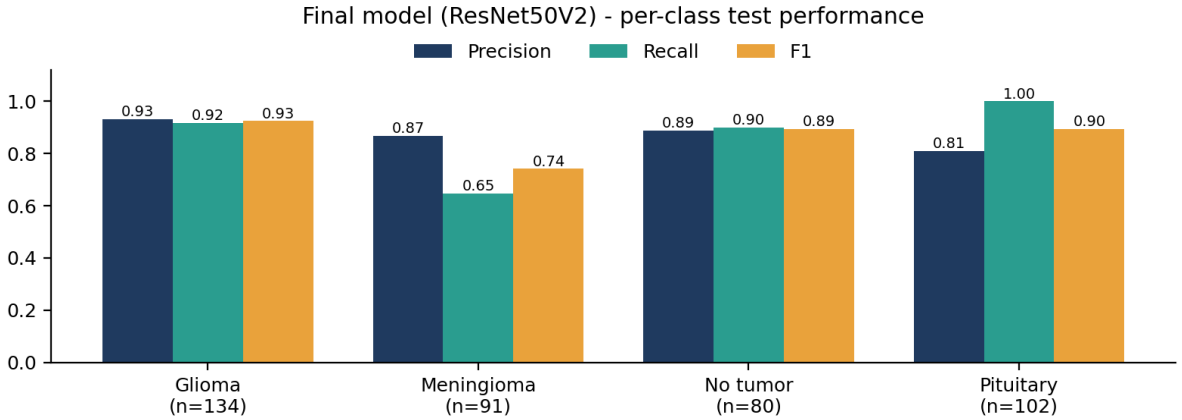


Figure 6 - Per-class precision, recall and F1 on the 407-image test set.

C1 Meningioma is the weak point	
Observation	Nearly one in three meningiomas is missed.
Evidence	Recall 0.648 (59 of 91 found, 32 missed), F1 0.742. The same ~65% recall appears for MobileNetV2 baseline, and EfficientNetB0 baseline falls to 0.451, so the difficulty is consistent across architectures.
Implication	Meningioma versus glioma/pituitary confusion deserves targeted work: more meningioma data, per-class thresholds, and review of the misclassified images.

C2 Pituitary: perfect recall, modest precision	
Observation	No pituitary tumor is missed, but other classes are labelled pituitary.
Evidence	Recall 1.000, precision 0.810. Implied false positives are about 24 (102 / 0.810 = ~126 predicted; derived from the reported precision).
Implication	The model leans toward pituitary when uncertain. This is a safe direction for sensitivity but adds follow-up workload; calibration would help.

C3 Tumor-versus-normal screening looks stronger than tumor typing	
Observation	Most errors are between tumor types, not between tumor and normal.
Evidence	No-tumor precision 0.889 on about 81 predicted no-tumor scans implies roughly 9 tumor scans labelled normal out of 327 (~97% tumor-detection sensitivity; derived from precision and support, so confirm with the exact confusion matrix).
Implication	The system is better suited to prioritising which scans a radiologist sees first than to naming the tumor type unaided.

D. Deployment and process insights

- **Explainability is built in.** The Streamlit app shows prediction, confidence gauge, class probabilities, margin, latency, a test-time-augmentation stability check and Grad-CAM for any class, plus a session log with CSV export.
- **Model hosting.** The 101 MB best_model.h5 is fetched from a Hugging Face Hub repository, keeping the GitHub repo light; convert_model.py re-saves it in native .keras format for version compatibility.
- **Statistical rigour.** Three hypothesis tests, grouped cross-validation and a reload sanity check (100% prediction agreement, zero probability difference) support reproducibility.
- **Single-split noise.** With 80 no-tumor test images, one error moves recall by about 1 pp, so small leaderboard gaps (e.g. 0.855 vs 0.864) should not be over-read.

E. Prioritised recommendations

#	Action	Why it matters
1	Review Grad-CAM overlays for correct and wrong predictions; confirm attention sits on tissue, not background or borders	Rules out the brightness / framing shortcut flagged in D3
2	Tune ResNet50V2 and compare it fairly with tuned MobileNetV2	Final choice currently compares a baseline against tuned models
3	Add targeted meningioma improvements (data, class-specific thresholds, error review)	Largest single weakness (recall 0.648)
4	Calibrate probabilities and add an 'uncertain - refer' band	Turns confidence into a usable triage signal
5	Validate externally on multi-centre data with patient-level splits	Dataset has no patient IDs; clinical claims are impossible without it

Housekeeping note: reports/model_comparison.json and the cnn_* / efficientnet_* images in the reports folder reflect an earlier run (CNN accuracy 19.9%, EfficientNet 28.9%) and disagree with the final leaderboard in artifacts/leaderboard.csv. This document uses the leaderboard and notebook outputs; regenerate or remove the stale files before publishing the repository.